

Figure 1: A 2R planar robot moving on a horizontal plane.

- Derive the inertia matrix $B(q)$ of a planar 2R robot using the *absolute* coordinates $q = (q_1, q_2)$ defined in Fig. 1. Will Coriolis and/or centrifugal terms be present in the quadratic velocity term $c(q, \dot{q})$ of the dynamic model? Why? In the absence of dissipative effects, what is the relation between the input torques u acting in this case on the right-hand side of the Euler-Lagrange equations and the torques $u_\theta = (u_{\theta 1}, u_{\theta 2})$ produced by two motors, one for each joint, directly connected to the respective joint axes?

$$p_{c1} = \begin{cases} d_1 c_1 \\ d_1 s_1 \end{cases} \quad p_{c2} = \begin{cases} l_1 c_1 + d_2 c_2 \\ l_1 s_1 + d_2 s_2 \end{cases}$$

$$v_{c1}^T v_{c1} = d_1^2 \dot{q}_1^2 \quad \omega_1 = \dot{q}_1$$

$$v_{c2}^T v_{c2} = l_1^2 \dot{q}_1^2 + d_2^2 \dot{q}_2^2 + 2 l_1 d_2 \cos(q_1 - q_2) \dot{q}_1 \dot{q}_2$$

$$v_{c1} = \begin{cases} -d_1 s_1 \dot{q}_1 \\ d_1 c_1 \dot{q}_1 \end{cases} \quad v_{c2} = \begin{cases} -l_1 s_1 \dot{q}_1 - d_2 s_2 \dot{q}_2 \\ l_1 c_1 \dot{q}_1 + d_2 c_2 \dot{q}_2 \end{cases}$$

$$T = \frac{1}{2} (I_1 + d_1^2 m_1) \dot{q}_1^2 + \frac{1}{2} I_2 \dot{q}_2^2 + \frac{1}{2} m_2 (l_1^2 \dot{q}_1^2 + d_2^2 \dot{q}_2^2 + 2 l_1 d_2 \cos(q_1 - q_2) \dot{q}_1 \dot{q}_2)$$

$$M_{11} = I_1 + d_1^2 m_1 + l_1^2 m_2 = e_1 \quad M_{12} = m_2 l_1 d_2 \overbrace{\cos(q_1 - q_2)}^{C_{1-2}} = e_2 C_{1-2}$$

$$M_{22} = I_2 + m_2 d_2^2 = e_3$$

$$M(q) = \begin{pmatrix} e_1 & e_2 C_{1-2} \\ e_2 C_{1-2} & e_3 \end{pmatrix}$$

$$\frac{\partial C_{1-2}}{\partial q_1} = -S_{1-2} = -S_{21} \quad \frac{\partial C_{1-2}}{\partial q_2} = S_{1-2} = S_{21} \quad \begin{matrix} \nearrow \text{notation} \\ \searrow \end{matrix}$$

$$C_1 = \frac{1}{2} \left(\begin{pmatrix} 0 & -e_2 s_{21} \\ -e_2 s_{21} & 2e_2 s_{21} \end{pmatrix} - \begin{pmatrix} 0 & -e_2 s_{21} \\ -e_2 s_{21} & 0 \end{pmatrix} \right) = e_2 s_{21} \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \Rightarrow C_1 = e_2 \sin(q_1 - q_2) \dot{q}_2^2$$

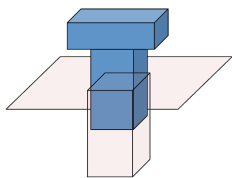
$$C_2 = \frac{1}{2} \left(\begin{pmatrix} -2e_2 s_{21} & e_2 s_{21} \\ e_2 s_{21} & 0 \end{pmatrix} - \begin{pmatrix} 0 & e_2 s_{21} \\ e_2 s_{21} & 0 \end{pmatrix} \right) = -e_2 s_{21} \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \Rightarrow C_2 = -e_2 s_{21} \dot{q}_1^2$$

The motors produces torques on

$$S(q) \cdot p = \begin{cases} p_1 = q_1 \\ p_2 = q_2 - q_1 \end{cases}$$

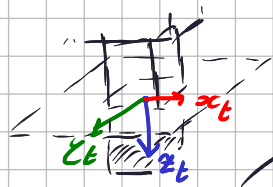
$$\dot{p} = \underbrace{\begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix}}_{J_p = \frac{\partial S}{\partial q}} \dot{q} \Rightarrow u_q = J^T u_p \xrightarrow{\text{exercise notation}} u = J^T u_\theta$$

$$\begin{cases} u_1 = u_{\theta 1} - u_{\theta 2} \\ u_2 = u_{\theta 2} \end{cases}$$



For the task of inserting a peg with a square section into a square hole, as depicted in Fig. 2, define i) a suitable *task frame*, ii) the *natural constraints* imposed by the geometry of the rigid and frictionless environment on the generalized (i.e., linear and angular) motion/force quantities expressed in this task frame, and iii) the *virtual constraints* that can be taken as reference values by a hybrid force-velocity control law for the smooth execution of the task.

We set a task frame RF_t with x_t, y_t forming a plane parallel to the surface on which the hole lies, and z_t in the direction of insertion. the natural constraints are 5:



the object can only be moved in z_t

$$v_y = v_x = 0 \quad \omega_x = \omega_y = \omega_z = 0 \quad \Rightarrow \quad s_z = 0$$

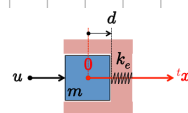
Assuming that the peg is already inside the square section. The virtual constraints are: $s_x = s_y = \mu_x = \mu_y = \mu_z = 0$ to not stress the peg. Then $v_z = v_{z,d}(t) \Rightarrow$ reference for the control.

Exercise 3

For the peg-in-hole task considered in Exercise 2, consider the problem of regulating the contact force in one direction against a side of the hole. The hole has now a compliant behavior, as modeled by a spring of stiffness $k_e > 0$. With reference to Fig. 3, which displays a horizontal section of the peg in contact with one side of the hole, let $m > 0$ be the mass of the peg and $d > 0$ the position of the undeformed contact along the x direction.

Design two controllers for the scalar command u such that the contact force F_c is regulated to a desired value $F_d > 0$, with an asymptotically stable transient behavior. In particular:

- suppose first that no force sensing is available and design a *compliance control law*;
- introduce an ideal force sensor (e.g., on the peg surface) and design a *force control law*;
- discuss robustness of the two designs w.r.t. uncertainties in the knowledge of k_e , d and m .



let x to be the position of the contact, $x = d = \text{rest}$ so

$$F_c = k_e(d - x) \quad \text{The dynamic model is} \quad m\ddot{x} = F_c + u$$

$$\text{Assume } F_c \text{ is measurable, we want } F_d = -k_e(d - x) \Rightarrow x_d = d + \frac{F_d}{k_e}$$

$$m\ddot{x} = k_e(d - x) + u \quad \text{we let } u = K_p(F_d - F_c) + F_d \quad (\text{PID} + \text{SSW})$$

$$m\ddot{x} = F_c - K_p F_d - K_p F_c + F_d \quad k_e(d - x) = F_c \Rightarrow x = d - \frac{F_c}{k_e}$$

$$m\ddot{x} = F_c(1 - K_p) + F_d(1 - K_p)$$

$$m\left(d - \frac{F_c}{k_e}\right)s^2 = F_c(1 - K_p) + F_d(1 - K_p)$$

$$m d s^2 = F_c\left(1 - K_p + \frac{m}{k_e} s^2\right) + F_d(1 - K_p)$$

$$F_c = \frac{m d s^2 + F_d(1 + K_p)}{(1 - K_p + \frac{m}{k_e} s^2)}, \quad \lim_{s \rightarrow 0} F_c = \lim_{t \rightarrow \infty} F_c = F_d$$